

IIT JAM 2021 Official Paper

Category: CBT Era (2015–2026) • Official Mock Test Series

TOTAL QUESTIONS

60

TOTAL MARKS

100 M

TEST DURATION

180 Min

PAPER PATTERN

MCQ • MSQ • NAT
(60 Ques)

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	30	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	10	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	20	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2021 • Q1 • ID: JAM_2021_Q1)

MCQ

+1 M

Neg: -0.33

Q. 1 Let $0 < \alpha < 1$ be a real number. The number of differentiable functions $y : [0, 1] \rightarrow [0, \infty)$, having continuous derivative on $[0, 1]$ and satisfying

$$\begin{aligned}y'(t) &= (y(t))^\alpha, \quad t \in [0, 1], \\y(0) &= 0,\end{aligned}$$

is

(A) exactly one.

(B) exactly two.

(C) finite but more than two.

(D) infinite.

Question 2 (JAM 2021 • Q2 • ID: JAM_2021_Q2)

MCQ

+1 M

Neg: -0.33

Q. 2 Let $P : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that $P(x) > 0$ for all $x \in \mathbb{R}$. Let y be a twice differentiable function on $\mathbb{R}$ satisfying $y''(x) + P(x)y'(x) - y(x) = 0$ for all $x \in \mathbb{R}$. Suppose that there exist two real numbers a, b ($a < b$) such that $y(a) = y(b) = 0$. Then

(A) $y(x) = 0$ for all $x \in [a, b]$.

(B) $y(x) > 0$ for all $x \in (a, b)$.

(C) $y(x) < 0$ for all $x \in (a, b)$.

(D) $y(x)$ changes sign on (a, b) .

Question 3 (JAM 2021 • Q3 • ID: JAM_2021_Q3)

MCQ

+1 M

Neg: -0.33

Q. 3 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function satisfying $f(x) = f(x+1)$ for all $x \in \mathbb{R}$. Then

- (A) f is not necessarily bounded above.
- (B) there exists a unique $x_0 \in \mathbb{R}$ such that $f(x_0 + \pi) = f(x_0)$.
- (C) there is no $x_0 \in \mathbb{R}$ such that $f(x_0 + \pi) = f(x_0)$.
- (D) there exist infinitely many $x_0 \in \mathbb{R}$ such that $f(x_0 + \pi) = f(x_0)$.



MA

1 / 17

Question 4 (JAM 2021 • Q4 • ID: JAM_2021_Q4)

MCQ

+1 M

Neg: -0.33

JAM 2021

MATHEMATICS - MA

Q. 4 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that for all $x \in \mathbb{R}$,

$$\int_0^1 f(xt) dt = 0. \quad (*)$$

Then

- (A) f must be identically 0 on the whole of $\mathbb{R}$.
- (B) there is an f satisfying (*) that is identically 0 on $(0, 1)$ but not identically 0 on the whole of $\mathbb{R}$.
- (C) there is an f satisfying (*) that takes both positive and negative values.
- (D) there is an f satisfying (*) that is 0 at infinitely many points, but is not identically zero.

Question 5 (JAM 2021 • Q5 • ID: JAM_2021_Q5)

MCQ

+1 M

Neg: -0.33

Q. 5 Let p and t be positive real numbers. Let D_t be the closed disc of radius t centered at $(0, 0)$, i.e., $D_t = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq t^2\}$. Define

$$I(p, t) = \iint_{D_t} \frac{dxdy}{(p^2 + x^2 + y^2)^p}.$$

Then $\lim_{t \rightarrow \infty} I(p, t)$ is finite

(A) only if $p > 1$.

(B) only if $p = 1$.

(C) only if $p < 1$.

(D) for no value of p .

Question 6 (JAM 2021 • Q6 • ID: JAM_2021_Q6)

MCQ

+1 M

Neg: -0.33

Q. 6 How many elements of the group $\mathbb{Z}_{50}$ have order 10?

(A) 10

(B) 4

(C) 5

(D) 8



MA

2 / 17

Question 7 (JAM 2021 • Q7 • ID: JAM_2021_Q7)

MCQ

+1 M

Neg: -0.33

JAM 2021

MATHEMATICS - MA

Q. 7 For every $n \in \mathbb{N}$, let $f_n : \mathbb{R} \rightarrow \mathbb{R}$ be a function. From the given choices, pick the statement that is the negation of

“For every $x \in \mathbb{R}$ and for every real number $\epsilon > 0$, there exists an integer $N > 0$ such that $\sum_{i=1}^p |f_{N+i}(x)| < \epsilon$ for every integer $p > 0$.”

- (A) For every $x \in \mathbb{R}$ and for every real number $\epsilon > 0$, there does not exist any integer $N > 0$ such that $\sum_{i=1}^p |f_{N+i}(x)| < \epsilon$ for every integer $p > 0$.
- (B) For every $x \in \mathbb{R}$ and for every real number $\epsilon > 0$, there exists an integer $N > 0$ such that $\sum_{i=1}^p |f_{N+i}(x)| \geq \epsilon$ for some integer $p > 0$.
- (C) There exists $x \in \mathbb{R}$ and there exists a real number $\epsilon > 0$ such that for every integer $N > 0$, there exists an integer $p > 0$ for which the inequality $\sum_{i=1}^p |f_{N+i}(x)| \geq \epsilon$ holds.
- (D) There exists $x \in \mathbb{R}$ and there exists a real number $\epsilon > 0$ such that for every integer $N > 0$ and for every integer $p > 0$ the inequality $\sum_{i=1}^p |f_{N+i}(x)| \geq \epsilon$ holds.

Question 8 (JAM 2021 • Q8 • ID: JAM_2021_Q8)

MCQ

+1 M

Neg: -0.33

Q. 8 Which one of the following subsets of $\mathbb{R}$ has a non-empty interior?

- (A) The set of all irrational numbers in $\mathbb{R}$.
- (B) The set $\{a \in \mathbb{R} : \sin(a) = 1\}$.
- (C) The set $\{b \in \mathbb{R} : x^2 + bx + 1 = 0 \text{ has distinct roots}\}$.
- (D) The set of all rational numbers in $\mathbb{R}$.

Question 9 (JAM 2021 • Q9 • ID: JAM_2021_Q9)

MCQ

+1 M

Neg: -0.33

Q. 9 For an integer $k \geq 0$, let P_k denote the vector space of all real polynomials in one variable of degree less than or equal to k . Define a linear transformation $T : P_2 \rightarrow P_3$ by

$$Tf(x) = f''(x) + xf(x).$$

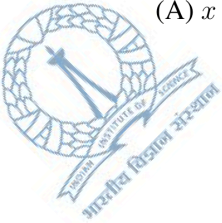
Which one of the following polynomials is not in the range of T ?

(A) $x + x^2$

(B) $x^2 + x^3 + 2$

(C) $x + x^3 + 2$

(D) $x + 1$



MA

3 / 17

Question 10 (JAM 2021 • Q10 • ID: JAM_2021_Q10)

MCQ

+1 M

Neg: -0.33

JAM 2021

MATHEMATICS - MA

Q. 10 Let $n > 1$ be an integer. Consider the following two statements for an arbitrary $n \times n$ matrix A with complex entries.

- I. If $A^k = I_n$ for some integer $k \geq 1$, then all the eigenvalues of A are k^{th} roots of unity.
II. If, for some integer $k \geq 1$, all the eigenvalues of A are k^{th} roots of unity, then $A^k = I_n$.

Then

- (A) both I and II are TRUE. (B) I is TRUE but II is FALSE.
(C) I is FALSE but II is TRUE. (D) neither I nor II is TRUE.



MA

4 / 17

Question 11 (JAM 2021 • Q11 • ID: JAM_2021_Q11)

MCQ

+2 M

Neg: -0.67

Q. 11 – Q. 30 carry two marks each.

Q. 11 Let $M_n(\mathbb{R})$ be the real vector space of all $n \times n$ matrices with real entries, $n \geq 2$.Let $A \in M_n(\mathbb{R})$. Consider the subspace W of $M_n(\mathbb{R})$ spanned by $\{I_n, A, A^2, \dots\}$. Then the dimension of W over $\mathbb{R}$ is necessarily(A) ∞ .(B) n^2 .(C) n .(D) at most n .

Question 12 (JAM 2021 • Q12 • ID: JAM_2021_Q12)

MCQ

+2 M

Neg: -0.67

Q. 12 Let y be the solution of

$$(1+x)y''(x) + y'(x) - \frac{1}{1+x}y(x) = 0, \quad x \in (-1, \infty),$$

$$y(0) = 1, \quad y'(0) = 0.$$

Then

(A) y is bounded on $(0, \infty)$.(B) y is bounded on $(-1, 0]$.(C) $y(x) \geq 2$ on $(-1, \infty)$.(D) y attains its minimum at $x = 0$.

Question 13 (JAM 2021 • Q13 • ID: JAM_2021_Q13)

MCQ

+2 M

Neg: -0.67

Q. 13 Consider the surface $S = \{(x, y, xy) \in \mathbb{R}^3 : x^2 + y^2 \leq 1\}$. Let $\vec{F} = y\hat{i} + x\hat{j} + \hat{k}$. If $\hat{n}$ is the continuous unit normal field to the surface S with positive z -component, then

$$\iint_S \vec{F} \cdot \hat{n} \, dS$$

equals

(A) $\frac{\pi}{4}$.(B) $\frac{\pi}{2}$.(C) π .(D) 2π .

Question 14 (JAM 2021 • Q14 • ID: JAM_2021_Q14)

MCQ

+2 M

Neg: -0.67

Q. 14 Consider the following statements.

- I. The group $(\mathbb{Q}, +)$ has no proper subgroup of finite index.
 II. The group $(\mathbb{C} \setminus \{0\}, \cdot)$ has no proper subgroup of finite index.

Which one of the following statements is true?

- (A) Both I and II are TRUE. (B) I is TRUE but II is FALSE.
 (C) II is TRUE but I is FALSE. (D) Neither I nor II is TRUE.

MA

5 / 17

Question 15 (JAM 2021 • Q15 • ID: JAM_2021_Q15)

MCQ

+2 M

Neg: -0.67

JAM 2021

MATHEMATICS - MA

Q. 15 Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a bijective map such that

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^2} < +\infty.$$

The number of such bijective maps is

- (A) exactly one. (B) zero.
 (C) finite but more than one. (D) infinite.

Question 16 (JAM 2021 • Q16 • ID: JAM_2021_Q16)

MCQ

+2 M

Neg: -0.67

Q. 16 Define

$$S = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \cdots \left(1 - \frac{1}{n^2}\right).$$

Then

- (A) $S = 1/2$. (B) $S = 1/4$. (C) $S = 1$. (D) $S = 3/4$.

Question 17 (JAM 2021 • Q17 • ID: JAM_2021_Q17)

MCQ

+2 M

Neg: -0.67

Q. 17 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an infinitely differentiable function such that for all $a, b \in \mathbb{R}$ with $a < b$,

$$\frac{f(b) - f(a)}{b - a} = f'\left(\frac{a + b}{2}\right).$$

Then

- (A) f must be a polynomial of degree less than or equal to 2.
- (B) f must be a polynomial of degree greater than 2.
- (C) f is not a polynomial.
- (D) f must be a linear polynomial.



MA

6 / 17

Question 18 (JAM 2021 • Q18 • ID: JAM_2021_Q18)

MCQ

+2 M

Neg: -0.67

JAM 2021

MATHEMATICS - MA

Q. 18 Consider the function

$$f(x) = \begin{cases} 1 & \text{if } x \in (\mathbb{R} \setminus \mathbb{Q}) \cup \{0\}, \\ 1 - \frac{1}{p} & \text{if } x = \frac{n}{p}, n \in \mathbb{Z} \setminus \{0\}, p \in \mathbb{N} \text{ and } \gcd(n, p) = 1. \end{cases}$$

Then

- (A) all $x \in \mathbb{Q} \setminus \{0\}$ are strict local minima for f .
- (B) f is continuous at all $x \in \mathbb{Q}$.
- (C) f is not continuous at all $x \in \mathbb{R} \setminus \mathbb{Q}$.
- (D) f is not continuous at $x = 0$.

Question 19 (JAM 2021 • Q19 • ID: JAM_2021_Q19)

MCQ

+2 M

Neg: -0.67

Q. 19 Consider the family of curves $x^2 - y^2 = ky$ with parameter $k \in \mathbb{R}$. The equation of the orthogonal trajectory to this family passing through $(1, 1)$ is given by

(A) $x^3 + 3xy^2 = 4$.

(B) $x^2 + 2xy = 3$.

(C) $y^2 + 2x^2y = 3$.

(D) $x^3 + 2xy^2 = 3$.

Question 20 (JAM 2021 • Q20 • ID: JAM_2021_Q20)

MCQ

+2 M

Neg: -0.67

Q. 20 Which one of the following statements is true?

(A) Exactly half of the elements in any even order subgroup of S_5 must be even permutations.(B) Any abelian subgroup of S_5 is trivial.(C) There exists a cyclic subgroup of S_5 of order 6.(D) There exists a normal subgroup of S_5 of index 7.

Question 21 (JAM 2021 • Q21 • ID: JAM_2021_Q21)

MCQ

+2 M

Neg: -0.67

Q. 21 Let $f : [0, 1] \rightarrow [0, \infty)$ be a continuous function such that

$$(f(t))^2 < 1 + 2 \int_0^t f(s) ds, \text{ for all } t \in [0, 1].$$

Then

(A) $f(t) < 1 + t$ for all $t \in [0, 1]$.(B) $f(t) > 1 + t$ for all $t \in [0, 1]$.(C) $f(t) = 1 + t$ for all $t \in [0, 1]$.(D) $f(t) < 1 + \frac{t}{2}$ for all $t \in [0, 1]$.

Question 22 (JAM 2021 • Q22 • ID: JAM_2021_Q22)

MCQ

+2 M

Neg: -0.67

Q. 22 Let A be an $n \times n$ invertible matrix and C be an $n \times n$ nilpotent matrix. If $X = \begin{pmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{pmatrix}$ is a $2n \times 2n$ matrix (each X_{ij} being $n \times n$) that commutes with the $2n \times 2n$ matrix $B = \begin{pmatrix} A & 0 \\ 0 & C \end{pmatrix}$, then

- (A) X_{11} and X_{22} are necessarily zero matrices.
- (B) X_{12} and X_{21} are necessarily zero matrices.
- (C) X_{11} and X_{21} are necessarily zero matrices.
- (D) X_{12} and X_{22} are necessarily zero matrices.

Question 23 (JAM 2021 • Q23 • ID: JAM_2021_Q23)

MCQ

+2 M

Neg: -0.67

Q. 23 Let $D \subseteq \mathbb{R}^2$ be defined by $D = \mathbb{R}^2 \setminus \{(x, 0) : x \in \mathbb{R}\}$. Consider the function $f : D \rightarrow \mathbb{R}$ defined by

$$f(x, y) = x \sin \frac{1}{y}.$$

Then

- (A) f is a discontinuous function on D .
- (B) f is a continuous function on D and cannot be extended continuously to any point outside D .
- (C) f is a continuous function on D and can be extended continuously to $D \cup \{(0, 0)\}$.
- (D) f is a continuous function on D and can be extended continuously to the whole of $\mathbb{R}^2$.

Question 24 (JAM 2021 • Q24 • ID: JAM_2021_Q24)

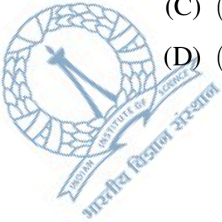
MCQ

+2 M

Neg: -0.67

Q. 24 Which one of the following statements is true?

- (A) $(\mathbb{Z}, +)$ is isomorphic to $(\mathbb{R}, +)$.
 (B) $(\mathbb{Z}, +)$ is isomorphic to $(\mathbb{Q}, +)$.
 (C) $(\mathbb{Q}/\mathbb{Z}, +)$ is isomorphic to $(\mathbb{Q}/2\mathbb{Z}, +)$.
 (D) $(\mathbb{Q}/\mathbb{Z}, +)$ is isomorphic to $(\mathbb{Q}, +)$.



MA

8 / 17

Question 25 (JAM 2021 • Q25 • ID: JAM_2021_Q25)

MCQ

+2 M

Neg: -0.67

JAM 2021

MATHEMATICS - MA

Q. 25 Let y be a twice differentiable function on $\mathbb{R}$ satisfying

$$y''(x) = 2 + e^{-|x|}, \quad x \in \mathbb{R},$$

$$y(0) = -1, \quad y'(0) = 0.$$

Then

- (A) $y = 0$ has exactly one root.
 (B) $y = 0$ has exactly two roots.
 (C) $y = 0$ has more than two roots.
 (D) there exists an $x_0 \in \mathbb{R}$ such that $y(x_0) \geq y(x)$ for all $x \in \mathbb{R}$.

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Question 26 (JAM 2021 • Q26 • ID: JAM_2021_Q26)

MCQ

+2 M

Neg: -0.67

Q. 26 Let $f : [0, 1] \rightarrow [0, 1]$ be a non-constant continuous function such that $f \circ f = f$. Define

$$E_f = \{x \in [0, 1] : f(x) = x\}.$$

Then

- (A) E_f is neither open nor closed. (B) E_f is an interval.
 (C) E_f is empty. (D) E_f need not be an interval.

Question 27 (JAM 2021 • Q27 • ID: JAM_2021_Q27)

MCQ

+2 M

Neg: -0.67

Q. 27 Let g be an element of S_7 such that g commutes with the element $(2, 6, 4, 3)$. The number of such g is

- (A) 6. (B) 4. (C) 24. (D) 48.

Question 28 (JAM 2021 • Q28 • ID: JAM_2021_Q28)

MCQ

+2 M

Neg: -0.67

Q. 28 Let G be a finite abelian group of odd order. Consider the following two statements:

- I. The map $f : G \rightarrow G$ defined by $f(g) = g^2$ is a group isomorphism.
 II. The product $\prod_{g \in G} g = e$.

- (A) Both I and II are TRUE. (B) I is TRUE but II is FALSE.
 (C) II is TRUE but I is FALSE. (D) Neither I nor II is TRUE.

Question 29 (JAM 2021 • Q29 • ID: JAM_2021_Q29)

MCQ

+2 M

Neg: -0.67

JAM 2021

MATHEMATICS - MA

Q. 29 Let $n \geq 2$ be an integer. Let $A : \mathbb{C}^n \rightarrow \mathbb{C}^n$ be the linear transformation defined by

$$A(z_1, z_2, \dots, z_n) = (z_n, z_1, z_2, \dots, z_{n-1}).$$

Which one of the following statements is true for every $n \geq 2$?

(A) A is nilpotent.

(B) All eigenvalues of A are of modulus 1.

(C) Every eigenvalue of A is either 0 or 1.

(D) A is singular.

Question 30 (JAM 2021 • Q30 • ID: JAM_2021_Q30)

MCQ

+2 M

Neg: -0.67

Q. 30 Consider the two series

$$\text{I. } \sum_{n=1}^{\infty} \frac{1}{n^{1+(1/n)}} \quad \text{and} \quad \text{II. } \sum_{n=1}^{\infty} \frac{1}{n^{2-n^{1/n}}}$$

Which one of the following holds?

- (A) Both I and II converge. (B) Both I and II diverge.
(C) I converges and II diverges. (D) I diverges and II converges.



MA

10 / 17

Question 31 (JAM 2021 • Q31 • ID: JAM_2021_Q31)

MSQ

+2 M

Neg: Nil (0)

Q. 31 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function with the property that for every $y \in \mathbb{R}$, the value of the expression

$$\sup_{x \in \mathbb{R}} [xy - f(x)]$$

is finite. Define $g(y) = \sup_{x \in \mathbb{R}} [xy - f(x)]$ for $y \in \mathbb{R}$. Then

(A) g is even if f is even.

(B) f must satisfy $\lim_{|x| \rightarrow \infty} \frac{f(x)}{|x|} = +\infty$.

(C) g is odd if f is even.

(D) f must satisfy $\lim_{|x| \rightarrow \infty} \frac{f(x)}{|x|} = -\infty$.

Question 32 (JAM 2021 • Q32 • ID: JAM_2021_Q32)

MSQ

+2 M

Neg: Nil (0)

Q. 32 Consider the equation

$$x^{2021} + x^{2020} + \cdots + x - 1 = 0.$$

Then

(A) all real roots are positive.

(B) exactly one real root is positive.

(C) exactly one real root is negative.

(D) no real root is positive.

Question 33 (JAM 2021 • Q33 • ID: JAM_2021_Q33)

MSQ

+2 M

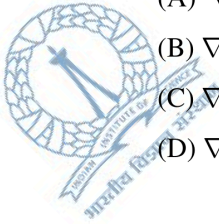
Neg: Nil (0)

Q. 33 Let $D = \mathbb{R}^2 \setminus \{(0, 0)\}$. Consider the two functions $u, v : D \rightarrow \mathbb{R}$ defined by

$$u(x, y) = x^2 - y^2 \text{ and } v(x, y) = xy.$$

Consider the gradients ∇u and ∇v of the functions u and v , respectively. Then

- (A) ∇u and ∇v are parallel at each point (x, y) of D .
 (B) ∇u and ∇v are perpendicular at each point (x, y) of D .
 (C) ∇u and ∇v do not exist at some points (x, y) of D .
 (D) ∇u and ∇v at each point (x, y) of D span $\mathbb{R}^2$.



MA

11 / 17

Question 34 (JAM 2021 • Q34 • ID: JAM_2021_Q34)

MSQ

+2 M

Neg: Nil (0)

JAM 2021

MATHEMATICS - MA

Q. 34 Consider the two functions $f(x, y) = x + y$ and $g(x, y) = xy - 16$ defined on $\mathbb{R}^2$. Then

- (A) the function f has no global extreme value subject to the condition $g = 0$.
 (B) the function f attains global extreme values at $(4, 4)$ and $(-4, -4)$ subject to the condition $g = 0$.
 (C) the function g has no global extreme value subject to the condition $f = 0$.
 (D) the function g has a global extreme value at $(0, 0)$ subject to the condition $f = 0$.

Question 35 (JAM 2021 • Q35 • ID: JAM_2021_Q35)

MSQ

+2 M

Neg: Nil (0)

Q. 35 Let $f : (a, b) \rightarrow \mathbb{R}$ be a differentiable function on (a, b) . Which of the following statements is/are true?

- (A) $f' > 0$ in (a, b) implies that f is increasing in (a, b) .
- (B) f is increasing in (a, b) implies that $f' > 0$ in (a, b) .
- (C) If $f'(x_0) > 0$ for some $x_0 \in (a, b)$, then there exists a $\delta > 0$ such that $f(x) > f(x_0)$ for all $x \in (x_0, x_0 + \delta)$.
- (D) If $f'(x_0) > 0$ for some $x_0 \in (a, b)$, then f is increasing in a neighbourhood of x_0 .

Question 36 (JAM 2021 • Q36 • ID: JAM_2021_Q36)

MSQ

+2 M

Neg: Nil (0)

Q. 36 Let G be a finite group of order 28. Assume that G contains a subgroup of order 7. Which of the following statements is/are true?

- (A) G contains a unique subgroup of order 7.
- (B) G contains a normal subgroup of order 7.
- (C) G contains no normal subgroup of order 7.
- (D) G contains at least two subgroups of order 7.

Question 37 (JAM 2021 • Q37 • ID: JAM_2021_Q37)

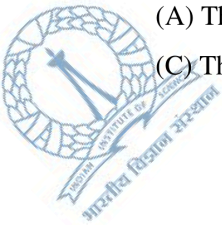
MSQ

+2 M

Neg: Nil (0)

Q. 37 Which of the following subsets of $\mathbb{R}$ is/are connected?

- (A) The set $\{x \in \mathbb{R} : x \text{ is irrational}\}$.
- (B) The set $\{x \in \mathbb{R} : x^3 - 1 \geq 0\}$.
- (C) The set $\{x \in \mathbb{R} : x^3 + x + 1 \geq 0\}$.
- (D) The set $\{x \in \mathbb{R} : x^3 - 2x + 1 \geq 0\}$.



Question 38 (JAM 2021 • Q38 • ID: JAM_2021_Q38)

MSQ

+2 M

Neg: Nil (0)

JAM 2021

MATHEMATICS - MA

Q. 38 Consider the four functions from $\mathbb{R}$ to $\mathbb{R}$:

$$f_1(x) = x^4 + 3x^3 + 7x + 1, \quad f_2(x) = x^3 + 3x^2 + 4x, \quad f_3(x) = \arctan(x)$$

and

$$f_4(x) = \begin{cases} x & \text{if } x \notin \mathbb{Z}, \\ 0 & \text{if } x \in \mathbb{Z}. \end{cases}$$

Which of the following subsets of $\mathbb{R}$ are open?(A) The range of f_1 .(B) The range of f_2 .(C) The range of f_3 .(D) The range of f_4 .

Question 39 (JAM 2021 • Q39 • ID: JAM_2021_Q39)

MSQ

+2 M

Neg: Nil (0)

Q. 39 Let V be a finite dimensional vector space and $T : V \rightarrow V$ be a linear transformation. Let $\mathcal{R}(T)$ denote the range of T and $\mathcal{N}(T)$ denote the null space $\{v \in V : Tv = 0\}$ of T . If $\text{rank}(T) = \text{rank}(T^2)$, then which of the following is/are necessarily true?

(A) $\mathcal{N}(T) = \mathcal{N}(T^2)$.(B) $\mathcal{R}(T) = \mathcal{R}(T^2)$.(C) $\mathcal{N}(T) \cap \mathcal{R}(T) = \{0\}$.(D) $\mathcal{N}(T) = \{0\}$.

Question 40 (JAM 2021 • Q40 • ID: JAM_2021_Q40)

MSQ

+2 M

Neg: Nil (0)

Q. 40 Let $m > 1$ and $n > 1$ be integers. Let A be an $m \times n$ matrix such that for some $m \times 1$ matrix b_1 , the equation $Ax = b_1$ has infinitely many solutions. Let b_2 denote an $m \times 1$ matrix different from b_1 . Then $Ax = b_2$ has

- (A) infinitely many solutions for some b_2 . (B) a unique solution for some b_2 .
(C) no solution for some b_2 . (D) finitely many solutions for some b_2 .



MA

13 / 17

Question 41 (JAM 2021 • Q41 • ID: JAM_2021_Q41)

NAT

+1 M

Neg: Nil (0)

Q. 41 The number of cycles of length 4 in S_6 is ____.

Question 42 (JAM 2021 • Q42 • ID: JAM_2021_Q42)

NAT

+1 M

Neg: Nil (0)

Q. 42 The value of

is ____.

$$\lim_{n \rightarrow \infty} \left(3^n + 5^n + 7^n \right)^{\frac{1}{n}}$$

Question 43 (JAM 2021 • Q43 • ID: JAM_2021_Q43)

NAT

+1 M

Neg: Nil (0)

Q. 43 Let $B = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 \leq 1\}$ and define $u(x, y, z) = \sin((1 - x^2 - y^2 - z^2)^2)$ for $(x, y, z) \in B$. Then the value of

$$\iiint_B \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) dx dy dz$$

is _____.

Question 44 (JAM 2021 • Q44 • ID: JAM_2021_Q44)

NAT

+1 M

Neg: Nil (0)

Q. 44 Consider the subset $S = \{(x, y) : x^2 + y^2 > 0\}$ of $\mathbb{R}^2$. Let

$$P(x, y) = \frac{y}{x^2 + y^2} \text{ and } Q(x, y) = -\frac{x}{x^2 + y^2}$$

for $(x, y) \in S$. If C denotes the unit circle traversed in the counter-clockwise direction, then the value of

$$\frac{1}{\pi} \int_C (P dx + Q dy)$$

is _____.

Question 45 (JAM 2021 • Q45 • ID: JAM_2021_Q45)

NAT

+1 M

Neg: Nil (0)

Q. 45 Consider the set $A = \{a \in \mathbb{R} : x^2 = a(a+1)(a+2) \text{ has a real root}\}$. The number of connected components of A is _____.

Question 46 (JAM 2021 • Q46 • ID: JAM_2021_Q46)

NAT

+1 M

Neg: Nil (0)

Q. 46 Let V be the real vector space of all continuous functions $f : [0, 2] \rightarrow \mathbb{R}$ such that the restriction of f to the interval $[0, 1]$ is a polynomial of degree less than or equal to 2, the restriction of f to the interval $[1, 2]$ is a polynomial of degree less than or equal to 3 and $f(0) = 0$. Then the dimension of V is equal to _____.

Question 47 (JAM 2021 • Q47 • ID: JAM_2021_Q47)

NAT

+1 M

Neg: Nil (0)

JAM 2021

MATHEMATICS - MA

Q. 47 The number of group homomorphisms from the group $\mathbb{Z}_4$ to the group S_3 is ____.

Question 48 (JAM 2021 • Q48 • ID: JAM_2021_Q48)

NAT

+1 M

Neg: Nil (0)

Q. 48 Let $y : (\frac{9}{10}, 3) \rightarrow \mathbb{R}$ be a differentiable function satisfying

$$(x - 2y)\frac{dy}{dx} + (2x + y) = 0, \quad x \in \left(\frac{9}{10}, 3\right), \quad \text{and } y(1) = 1.$$

Then $y(2)$ equals ____.

Question 49 (JAM 2021 • Q49 • ID: JAM_2021_Q49)

NAT

+1 M

Neg: Nil (0)

Q. 49 Let $\vec{F} = (y + 1)e^y \cos(x)\hat{i} + (y + 2)e^y \sin(x)\hat{j}$ be a vector field in $\mathbb{R}^2$ and C be a continuously differentiable path with the starting point $(0, 1)$ and the end point $(\frac{\pi}{2}, 0)$. Then

$$\int_C \vec{F} \cdot d\vec{r}$$

equals ____.

Question 50 (JAM 2021 • Q50 • ID: JAM_2021_Q50)

NAT

+1 M

Neg: Nil (0)

Q. 50 The value of

$$\frac{\pi}{2} \lim_{n \rightarrow \infty} \cos\left(\frac{\pi}{4}\right) \cos\left(\frac{\pi}{8}\right) \cdots \cos\left(\frac{\pi}{2^{n+1}}\right)$$

is _____.



MA

15 / 17

Question 51 (JAM 2021 • Q51 • ID: JAM_2021_Q51)

NAT

+2 M

Neg: Nil (0)

Q. 51 The number of elements of order two in the group S_4 is equal to _____.

Question 52 (JAM 2021 • Q52 • ID: JAM_2021_Q52)

NAT

+2 M

Neg: Nil (0)

Q. 52 The least possible value of k , accurate up to two decimal places, for which the following problem

$$\begin{aligned} y''(t) + 2y'(t) + ky(t) &= 0, t \in \mathbb{R}, \\ y(0) &= 0, y(1) = 0, y(1/2) = 1, \end{aligned}$$

has a solution is _____.

Question 53 (JAM 2021 • Q53 • ID: JAM_2021_Q53)

NAT

+2 M

Neg: Nil (0)

Q. 53 Consider those continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ that have the property that given any $x \in \mathbb{R}$,

$$f(x) \in \mathbb{Q} \text{ if and only if } f(x+1) \in \mathbb{R} \setminus \mathbb{Q}.$$

The number of such functions is ____.

Question 54 (JAM 2021 • Q54 • ID: JAM_2021_Q54)

NAT

+2 M

Neg: Nil (0)

Q. 54 The largest positive number a such that

$$\int_0^5 f(x) dx + \int_0^3 f^{-1}(x) dx \geq a$$

for every strictly increasing surjective continuous function $f : [0, \infty) \rightarrow [0, \infty)$ is ____.

Question 55 (JAM 2021 • Q55 • ID: JAM_2021_Q55)

NAT

+2 M

Neg: Nil (0)

Q. 55 Define the sequence

$$s_n = \begin{cases} \frac{1}{2^n} \sum_{j=0}^{n-2} 2^{2j} & \text{if } n > 0 \text{ is even,} \\ \frac{1}{2^n} \sum_{j=0}^{n-1} 2^{2j} & \text{if } n > 0 \text{ is odd.} \end{cases}$$

Define $\sigma_m = \frac{1}{m} \sum_{n=1}^m s_n$. The number of limit points of the sequence $\{\sigma_m\}$ is ____.



Question 56 (JAM 2021 • Q56 • ID: JAM_2021_Q56)

NAT

+2 M

Neg: Nil (0)

JAM 2021

MATHEMATICS - MA

Q. 56 The determinant of the matrix

$$\begin{pmatrix} 2021 & 2020 & 2020 & 2020 \\ 2021 & 2021 & 2020 & 2020 \\ 2021 & 2021 & 2021 & 2020 \\ 2021 & 2021 & 2021 & 2021 \end{pmatrix}$$

is ____.

Question 57 (JAM 2021 • Q57 • ID: JAM_2021_Q57)

NAT

+2 M

Neg: Nil (0)

Q. 57 The value of

$$\lim_{n \rightarrow \infty} \int_0^1 e^{x^2} \sin(nx) dx$$

is ____.

Question 58 (JAM 2021 • Q58 • ID: JAM_2021_Q58)

NAT

+2 M

Neg: Nil (0)

Q. 58 Let S be the surface defined by

$$\{(x, y, z) \in \mathbb{R}^3 : z = 1 - x^2 - y^2, z \geq 0\}.$$

Let $\vec{F} = -y\hat{i} + (x-1)\hat{j} + z^2\hat{k}$ and $\hat{n}$ be the continuous unit normal field to the surface S with positive z -component. Then the value of

$$\frac{1}{\pi} \iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS$$

is ____.

Question 59 (JAM 2021 • Q59 • ID: JAM_2021_Q59)**NAT****+2 M****Neg: Nil (0)**

Q. 59 Let $A = \begin{pmatrix} 2 & -1 & 3 \\ 2 & -1 & 3 \\ 3 & 2 & -1 \end{pmatrix}$. Then the largest eigenvalue of A is ____.

Question 60 (JAM 2021 • Q60 • ID: JAM_2021_Q60)**NAT****+2 M****Neg: Nil (0)**

Q. 60 Let $A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$. Consider the linear map T_A from the real vector space $M_4(\mathbb{R})$ to itself defined by $T_A(X) = AX - XA$, for all $X \in M_4(\mathbb{R})$. The dimension of the range of T_A is ____.

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

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Q. No.	Type	Marks	Official Key	Q. No.	Type	Marks	Official Key	Q. No.	Type	Marks	Official Key
1	MCQ	+1	D	21	MCQ	+2	A	41	NAT	+1	90
2	MCQ	+1	A	22	MCQ	+2	B	42	NAT	+1	7
3	MCQ	+1	D	23	MCQ	+2	C	43	NAT	+1	0
4	MCQ	+1	A	24	MCQ	+2	C	44	NAT	+1	-2
5	MCQ	+1	A	25	MCQ	+2	B	45	NAT	+1	2
6	MCQ	+1	B	26	MCQ	+2	B	46	NAT	+1	5
7	MCQ	+1	C	27	MCQ	+2	C	47	NAT	+1	4
8	MCQ	+1	C	28	MCQ	+2	A	48	NAT	+1	3
9	MCQ	+1	D	29	MCQ	+2	B	49	NAT	+1	1
10	MCQ	+1	B	30	MCQ	+2	B	50	NAT	+1	1
11	MCQ	+2	D	31	MSQ	+2	A, B	51	NAT	+2	9
12	MCQ	+2	D	32	MSQ	+2	A, B	52	NAT	+2	10.8 to 10.9
13	MCQ	+2	B	33	MSQ	+2	B, D	53	NAT	+2	0
14	MCQ	+2	A	34	MSQ	+2	A, D	54	NAT	+2	15
15	MCQ	+2	B	35	MSQ	+2	A, C	55	NAT	+2	0
16	MCQ	+2	A	36	MSQ	+2	A, B	56	NAT	+2	2021
17	MCQ	+2	A	37	MSQ	+2	B, C	57	NAT	+2	0
18	MCQ	+2	A	38	MSQ	+2	B, C, D	58	NAT	+2	2
19	MCQ	+2	A	39	MSQ	+2	A, B, C	59	NAT	+2	4
20	MCQ	+2	C	40	MSQ	+2	MTA	60	NAT	+2	8

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.